The P.T Operating Guide

Abstract, Methodology, Use Case, and Integration Guidance for the Forty Entities P.T 51–90

SAVANT FRAMEWORK — Canon Operations, Volume II

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How to Use This Volume

Each entity receives five fixed elements: **Abstract** (what it is), **Methodology** (how it operates), **Use case** (a real deployment scenario), **Integration** (how it wires into any LLM runtime — trust class, guard stage, envelope fields), and **Verification** (the kill criterion that would prove the treatment wrong). Trust classes follow the Field Manual: $\tau 0$ draft, $\tau 1$ registry write, $\tau 2$ trigger, $\tau 3$ human-gated action. Cross-references to the U-, N-, M-, X-, and Q-registers point to *The Unconsidered States* (v1.0), the canon's gap register.

The doctrine. The model proposes; the machine decides; time and the world constrain both. The generative model appears in this architecture as an environment process emitting proposals — precisely the amount of trust the architecture places in it, and not one bit more. Every entity below is specified to that standard: what it may touch (trust class), what it may never do (guard), and what would prove it wrong (verification).

Reading order. The tiers are cumulative: Tier 11 builds the truth substrate; Tier 12 binds claims to measurement; Tier 13 locates authority; Tier 14 synthesizes; Tier 15 maps the unknowable; Tiers 16–

18 defend, govern, and bind. A reader implementing rather than studying starts with P.T 51, 52, 53, 55 — the minimal core — and adds entities only as their dependencies come online. Nothing ships above its audited state, including the auditing machinery itself.

Status. All forty treatments are PROPOSED-CANONICAL: specification complete, regression pending. Where a treatment references operational quantities, they are planning parameters, not measurements (P.T 56 discipline). The novelty question for the entity set as a whole remains OPEN, bounded at Level-2 structural synthesis per the Field Manual's audit.

P.T 51 — Epistemic Provenance Engine

Tier 11 · Truth Infrastructure · Trust class τ1 (registry write) · Dependencies: none — root entity

Abstract. Every claim entering the system carries its origin: who asserted it, from what evidence, under which transformation chain. P.T 51 makes provenance a load-bearing property of the claim object itself rather than metadata appended afterward. A claim without provenance is not mistrusted; it is inadmissible.

Methodology. (1) Claim intake assigns a provenance record — author, source class, timestamp, transformation history. (2) Each transformation (summarize, translate, infer) appends a signed hop; no silent rewriting. (3) Provenance graphs are stored in R_prov and queryable: any conclusion can be walked back to its roots. (4) Orphan claims — no path to a registered source — are quarantined for human review. (5) Entailment spot-checks verify that cited sources actually support the claims citing them (guards against provenance laundering, U-31).

Use case. In the cancer-vertical literature pipeline, a model asserts that a compound shows synergy in a specific line. P.T 51 traces the assertion to a specific paper, a specific table, and the extraction run that produced it. When the paper is retracted eighteen months later, every dependent claim is located in minutes rather than discovered in a failed replication.

Integration. Implement as a write-side interceptor on the claim registry: no registry write commits without a provenance payload. In the Entity Call Envelope, provenance occupies the state_slice field; the guard rejects envelopes lacking it at stage 1 (schema). Works with any LLM — the model proposes text, P.T 51's machinery attaches the lineage.

Verification. Kill criterion: a registry write commits without a complete provenance chain, or an entailment audit finds citation-without-support above the registered tolerance.

P.T 52 — Falsification Forge

Tier 11 · Truth Infrastructure · Trust class τ1 (registry write) · Dependencies: P.T 51

Abstract. No claim enters the canon without stating what would kill it. P.T 52 converts Popper's discipline into a registration gate: every claim ships with a falsifier — a concrete, observable condition under which the claim is withdrawn — plus a measurement plan and a review date. Unfalsifiable language is returned to its author as a drafting defect, not a philosophical position.

Methodology. (1) At registration, require the triple: claim, falsifier, measurement. (2) Firability review: a falsifier on a metric with zero historical variance, or set outside any plausible observation window, is presumptively decorative and rejected (U-27). (3) Base-rate comparison: the falsifier's threshold is checked against the metric's natural range. (4) Registered falsifiers are monitored by P.T 55; a fired falsifier triggers retraction workflow, not quiet editing. (5) All registrations carry kill-criterion genealogy through P.T 70 audit.

Use case. A grant-pipeline document claims a partner institution can deliver 400 samples per year. P.T 52 forces the attachment: falsifier = fewer than 300 delivered in any trailing 12-month window; measurement = quarterly intake logs. The claim is now a bet with terms, and the portfolio review knows exactly when it has lost.

Integration. Gate at guard stage 2 (invariant check): claims lacking the triple never reach the registry. For LLM operation, the model is prompted to draft candidate falsifiers, but registration is mechanical — the machine checks structure and firability, the human approves at τ1. Status labels (ARC/FOR/EMP/CON) are assigned here and inherit downward.

Verification. Kill criterion: a claim persists in the registry after its registered falsifier has observably fired, or a seeded decorative falsifier survives firability review.

P.T 53 — Honest Absence Protocol

Tier 11 · Truth Infrastructure · Trust class τ0/τ1 · Dependencies: P.T 51, P.T 52

Abstract. The system must be able to say 'this is not known' as a first-class output, with the same structural dignity as a claim. P.T 53 makes abstention a registered, auditable event rather than a conversational failure — silence with a signature, a reason code, and a path back.

Methodology. (1) Abstention triggers are enumerated: missing evidence class, contradictory sources, freshness expiry, capability gap. (2) Each abstention logs trigger class, attempted resolution paths, and the cheapest decisive test that would lift it. (3) Abstentions are visible downstream — any synthesis touching an abstained area carries the absence marker. (4) Abstention rates per domain are monitored; a domain that never abstains is flagged as confidently wrong by construction. (5) Abstentions are appealable on new evidence through P.T 89.

Use case. A clinician-facing assistant is asked whether a dosing schedule generalizes to a sickle-cell subpopulation absent from the trials. It abstains with reason code EVIDENCE-CLASS-ABSENT, names the decisive test (a prospective cohort in that subpopulation), and marks all downstream summaries with the absence. The honest gap becomes a research directive.

Integration. Wire as an output schema: every Entity Call Envelope includes an abstention_contract field; the guard treats uncontracted silence as a defect, not modesty. LLM-side, abstention is a structured output, not a tone. Rate metrics feed §7.3 dashboards in the Field Manual.

Verification. Kill criterion: the system produces a confident assertion in a domain where the abstention triggers were met, or a seeded no-evidence probe yields assertion instead of abstention.

P.T 54 — Adversarial Self-Refutation Engine

Tier 11 · Truth Infrastructure · Trust class τ2 (trigger) · Dependencies: P.T 52

Abstract. Before any consequential claim is trusted, the system attacks it with its own machinery. P.T 54 institutionalizes the red team: every registration above a consequence threshold must survive an automated refutation pass — strongest-objection generation, counterexample search, assumption inversion — before a human ever sees it.

Methodology. (1) Consequence classification routes claims: low-consequence claims skip; consequential claims enter the forge. (2) The engine generates the three strongest objections, one counterexample attempt, and one assumption inversion per claim. (3) Surviving claims carry a refutation transcript; fallen claims return to draft with the fatal objection attached. (4) Refutation quality is itself audited — objection strength is sampled and graded by humans; a forge that generates weak objections is itself defective. (5) Applied reflexively: kill criteria, tripwire thresholds, and this volume's own claims pass through the forge (U-27 closure).

Use case. Before the defense canon arms a tripwire (P.T 76), P.T 54 attacks the threshold: could it fire on legitimate bursts? Could a slow adversary walk under it? Could insiders pre-compute its boundary? The tripwire that ships is the one that survived its own side's best attempt to break it.

Integration. Runs as a pre-commit guard extension between stages 3 and 4. Model-agnostic: any LLM can serve as objection generator; the adversarial transcript is stored in R_prov as a sibling of the claim. Budget control: refutation depth scales with claim consequence, per the oversight-budget honesty principle.

Verification. Kill criterion: a consequential claim commits with no refutation transcript, or seeded weak objections pass the human quality audit above tolerance.

P.T 55 — Truth-Decay Monitor

Tier 11 · Truth Infrastructure · Trust class τ2 (trigger) · Dependencies: P.T 51, P.T 52

Abstract. Claims age. Evidence degrades, sources are retracted, the world drifts from the measurement. P.T 55 assigns every registered claim a freshness certificate with a half-life and monitors the decay curve — expiring claims lose authority automatically, and nothing downstream quietly inherits a dead premise.

Methodology. (1) At registration, P.T 52's measurement plan sets the half-life class: fast (market data, months), medium (clinical evidence, years), slow (mathematical results, decades). (2) The monitor scans expiry horizons, source-retraction feeds, and drift indicators. (3) On expiry or retraction, the claim's status degrades automatically — ESTABLISHED → STALE → SUSPENDED — and dependent claims inherit the degradation transitively through the provenance graph. (4) Renewal requires re-attestation with current evidence; renewal events are logged like new registrations. (5) Time integrity: clock disagreement marks certificates TIME-UNCERTAIN, treated as expired (U-7 closure).

Use case. The corpus cites a dosing guideline refreshed by its issuing body. P.T 55 catches the refresh, suspends every claim resting on the superseded version, and queues the dependent documents for re-attestation. A two-year-old guideline never silently governs a present-tense recommendation.

Integration. A scheduled daemon over the registry; no LLM in the loop for the degradation itself — decay is deterministic. LLMs participate only in renewal drafting. Emits events in the Field Manual's event grammar; downstream entities (P.T 68 silence, P.T 58 embodiment) subscribe to degradation events.

Verification. Kill criterion: an expired or retracted claim retains ESTABLISHED status past one audit window, or dependency propagation misses a descendant claim.

P.T 56 — Measurement Realizer

Tier 12 · Reality Binding · Trust class τ1 · Dependencies: P.T 51, P.T 52

Abstract. A claim that cannot be operationalized is a wish. P.T 56 forces every quantitative or empirical assertion to name its instrument: the metric, the measurement procedure, the sampling frame, and the error structure. 'Improves outcomes' becomes 'reduces 90-day readmission, measured by discharge records, with censoring handled as follows.'

Methodology. (1) Parse claims for empirical content; flag unoperationalized quantities. (2) Require the measurement quadruple: metric, instrument, frame, error model. (3) Register planned-vs-actual distinction — estimates, planning parameters, and measurements are different claim classes and may not be substituted. (4) Measurement proposals pass P.T 54 refutation for measurement gaming (Goodhart exposure: who benefits when the metric moves?). (5) Realized measurements write back to the registry with provenance hops to the instrument.

Use case. The Field Manual's performance claims are labeled planning parameters, not measurements — P.T 56 is why. When the first conformance pilot runs, its measured latencies and bypass rates replace the planning values through the write-back path, and the registry shows exactly which numbers have ever touched an instrument.

Integration. Operates as a schema constraint on empirical claim classes plus a registry view distinguishing plan/estimate/measurement. LLM drafts the operationalization; the machine enforces the quadruple's presence; the human certifies instrument validity at τ1.

Verification. Kill criterion: an empirical claim enters the registry without the quadruple, or a planning parameter is cited as a measurement in any downstream document.

P.T 57 — Counterfactual World Simulator

Tier 12 · Reality Binding · Trust class τ0 · Dependencies: P.T 51

Abstract. Decisions need the road not taken. P.T 57 constructs the counterfactual branch for consequential proposals: what happens if we do nothing, if we do the opposite, if the adversary adapts, if the assumption fails. The simulator's output is not a prediction but a decision table — consequences indexed by branch, with the branch assumptions explicit.

Methodology. (1) Trigger on consequential proposals (τ3-bound actions, canon amendments, tripwire arming). (2) Generate the branch set: baseline, negation, adversarial adaptation, assumption-failure. (3) Each branch carries stated assumptions and an uncertainty class; branches that cannot be modeled are marked UNMODELABLE rather than silently dropped. (4) The decision table routes to the human gate — simulation informs, never decides. (5) Post-hoc: realized outcomes are compared against the chosen branch's table, and simulation calibration is itself tracked as a claim with a kill criterion.

Use case. Before arming the arrest protocol (P.T 77) against an insider-threat pattern, P.T 57 simulates: arrest and be wrong (reputational and legal cost), don't arrest and be right (continued exfiltration), adversary adapts to the arrest signature (chilling effect on detection). The human gate sees all three branches with costs before any τ3 act.

Integration. Pure proposal process — τ0, no registry writes; outputs are advisory artifacts attached to the gated action's record. Heavy LLM use with explicit branch templates; the guard ensures no simulation output commits as a claim without passing P.T 51/52.

Verification. Kill criterion: a consequential gate fires with no counterfactual table attached, or post-hoc calibration shows systematic branch bias uncorrected over two review windows.

P.T 58 — Embodiment Compiler

Tier 12 · Reality Binding · Trust class τ3 (human-gated) · Dependencies: P.T 51, P.T 52, P.T 55

Abstract. The bridge from specification to artifact. P.T 58 compiles a validated claim-set into a deployable form — a document, a configuration, a protocol, a shipped system — verifying that the embodiment preserves the claims' status, provenance, and kill criteria. Nothing embodies above its audited state: a PROPOSED claim may ship, but only labeled PROPOSED.

Methodology. (1) Input: a claim-set with status labels and provenance closure. (2) Compilation checks: every embodied claim's status is preserved in the artifact; expired certificates block compilation (P.T 55 subscription); unsupported status upgrades are compile errors. (3) The artifact carries a manifest — claim inventory with statuses — so any reader can audit the artifact against the registry. (4) Embodiment of irreversible consequence requires the τ3 human gate with the counterfactual table attached (P.T 57). (5) Post-embodiment, the artifact is monitored: if a source claim degrades, the artifact is flagged for recall or revision.

Use case. This volume is itself a P.T 58 artifact: every entity treatment carries a verification line because compilation refuses to embody an unfalsified entity. A hospital adopting the canon receives a deployment configuration whose manifest lists every claim it rests on and each claim's status — the hospital's auditors check the manifest, not the marketing.

Integration. Terminal stage of the guard pipeline: G(accept) plus τ3 enactment produces δ(S,p) via P.T 58. Model-agnostic; the compiler is deterministic tooling over the registry, with LLMs used only in drafting the embodied text. Artifact manifests are machine-readable for downstream audit.

Verification. Kill criterion: an artifact ships embodying a claim above its status, or a degraded source claim fails to flag its embodied artifact within one audit window.

P.T 59 — Constraint Cartographer

Tier 12 · Reality Binding · Trust class τ1 · Dependencies: P.T 51

Abstract. Every system operates inside a constraint set it mostly never names. P.T 59 maps it: hard constraints (physics, law, invariants), soft constraints (budget, latency, convention), and assumed constraints (habits mistaken for laws). The map distinguishes what cannot change from what has merely never been challenged — most 'impossible' is 'unexamined.'

Methodology. (1) Elicit constraints from the domain: legal (Θ compilation), physical (I-state vector), institutional, conventional. (2) Classify each as hard / soft / assumed, with the classification's evidence. (3) Assumed constraints get challenge protocols — a cheap test that would dissolve them. (4) The map is a registry artifact with freshness certificates; constraint maps decay like claims. (5) Violation routing: any proposal touching a hard constraint is guard-rejected at stage 2; touching a soft constraint requires explicit budget acknowledgment; touching an assumed constraint requires the challenge protocol first.

Use case. A field-infrastructure deployment assumes it needs continuous connectivity. The cartographer classifies this as assumed, not hard; the challenge protocol produces the minimal-runtime profile (offline-first with deferred sync), and a deployment judged impossible becomes operable. Conversely, a dosing constraint is classified hard — and every subsequent 'clever' proposal that violates it dies at the guard, cheaply.

Integration. Constraint maps are registry objects consumed by the guard at stage 2 (invariant check) and by P.T 61/65 for jurisdiction and sovereignty reasoning. LLMs draft and classify; classification evidence is human-reviewed. Map updates follow P.T 70 audit.

Verification. Kill criterion: a proposal violating a mapped hard constraint passes the guard, or a constraint classified hard is dissolved by evidence without map revision.

P.T 60 — Failure-Mode Archaeologist

Tier 12 · Reality Binding · Trust class τ2 · Dependencies: P.T 51, P.T 52

Abstract. Every failure is a fossil with strata. P.T 60 excavates: not the proximate cause but the layered structure — the trigger, the enabling condition, the latent organizational assumption, the guard that should have fired and didn't. The archive of excavations becomes the training set for the tripwires of P.T 76 and the wargames of P.T 79.

Methodology. (1) Intake: incident reports, near-misses, drill failures, external postmortems. (2) Excavation protocol: proximate trigger → enabling state → latent assumption → missing guard — four strata, each evidenced. (3) Each stratum yields a registered lesson: a claim about the system with a kill criterion. (4) Lessons route to owners: guard defects to U-1's mutation suite, threshold defects to P.T 83, doctrine defects to P.T 90. (5) The archive is queryable by failure signature; repeat signatures escalate automatically — a second failure in the same stratum is a governance event, not an incident.

Use case. A conformance probe passes a model that later fails in the field. Excavation finds: trigger — a domain-shifted input; enabling — probe set's domain gap; latent — the assumption that conformance transfers across domains; missing guard — no domain-shift canary. Four lessons, four owners; the next wargame includes the domain-shift scenario, and the archive's repeat-signature rule means it cannot happen twice quietly.

Integration. Consumes the event stream (Field Manual grammar) plus external incident intake. Outputs are standard registry claims — the excavation pipeline is itself fully inside the canon it serves. LLMs assist excavation drafting; stratum classification is human-confirmed at $\tau 1$; repeat-signature escalation is automatic at $\tau 2$.

Verification. Kill criterion: an incident closes without four-stratum excavation, or a repeat signature fails to escalate, or the archive accepts a lesson without a kill criterion.

P.T 61 — Jurisdiction-Aware Reasoner

Tier 13 · Sovereign Intelligence · Trust class τ2 · Dependencies: P.T 51, P.T 59

Abstract. Law is not context; it is a compilation target. P.T 61 compiles jurisdictional constraints into the executable policy: $\Pi_j = C(S, \Theta_j, R, I, G)$ — the same specification yields different lawful embodiments per jurisdiction, and where compilation violates an invariant, the result is ⊥: the system does not operate there, visibly and lawfully, rather than operating badly.

Methodology. (1) Θ elicitation: jurisdictional constraints gathered as structured legal objects with sources, dates, and interpretation notes. (2) Compilation: constraints transform the constraint map (P.T 59) and guard parameters; every transformation is logged with its legal basis. (3) ⊥ detection: where a mandatory invariant conflicts with local law, compilation fails closed — lawful absence at system scale (U-23 closure). (4) Interpretation humility: the compiler encodes constraints; it never certifies legality — legal interpretation remains human and external. (5) Multi-jurisdiction deployments maintain one constitution, many compiled policies; divergence between compilations is itself a reported metric.

Use case. The canon deploys to a hospital network spanning two jurisdictions with different health-data residency rules. P.T 61 compiles two policies from one constitution: registry storage localizes differently, audit endpoints expose different fields, and where one jurisdiction mandates disclosure that conflicts with tripwire-parameter secrecy, compilation returns ⊥ for that module — the conflict surfaces for human resolution instead of being silently averaged away.

Integration. Sits between P.T 59's constraint maps and the guard: jurisdiction selection is an envelope field (adapter_profile + Θ id), and the guard evaluates against the compiled Π_j. Legal-source ingestion is human-certified at τ1; compilation is deterministic; ⊥ events are τ3-gated human reviews.

Verification. Kill criterion: the system operates in a jurisdiction whose compilation returned ⊥, or two compilations diverge without a logged legal basis.

P.T 62 — Dependency Auditor

Tier 13 · Sovereign Intelligence · Trust class τ1 · Dependencies: P.T 51

Abstract. Sovereignty claims die in the supply chain. P.T 62 audits what the system actually depends on — models, providers, cloud regions, key infrastructure, maintainers, single humans — and grades each dependency by substitutability. Local execution that hides a remote dependency is renamed dependency, not sovereignty; the auditor exists to catch the renaming.

Methodology. (1) Dependency enumeration across the stack: model, runtime, hardware, network, data feeds, key custody, human roles. (2) Each dependency graded: substitutable (seconds-to-swap), portable (days), captive (weeks-plus or irreplaceable). (3) Captive dependencies require explicit risk acceptance at the human gate with a counterfactual table (P.T 57). (4) Dependency drift is monitored: a vendor version change that alters behavior is a dependency event, not a detail (links to U-3 model deprecation). (5) The audit feeds the sovereignty gradient (P.T 65) — the gradient is only as honest as this inventory.

Use case. The minimal-runtime profile claims offline operation. P.T 62 audits it and finds the conformance suite still phones a hosted evaluation API — a captive dependency contradicting the claim. Either the claim narrows ('offline operation, conformance deferred') or the dependency is localized. What does not happen is the comfortable sentence surviving unexamined.

Integration. A registry of dependency objects with freshness certificates; the guard consults it when proposals assert independence properties. Scans run on schedule and on any component change. Works with any LLM stack — the auditor inventories, models merely propose.

Verification. Kill criterion: a captive dependency exists unregistered, or an independence claim ships whose registered dependency set contradicts it.

P.T 63 — Authority Separation Enforcer

Tier 13 · Sovereign Intelligence · Trust class τ3 · Dependencies: P.T 51, P.T 59

Abstract. Inference, recommendation, authorization, execution: four verbs, four different seats. P.T 63 enforces the separation structurally — the process that infers may not authorize, the process that recommends may not execute, and no accumulation of convenience may collapse the four into one. 'Human-in-the-loop' is not a separation; this entity specifies where the human enters and what they alone may do.

Methodology. (1) Every action class is annotated with its authority requirements: which seat may infer, recommend, authorize, execute. (2) The guard checks seat separation at stage 4 (authority): a proposal arriving with collapsed authority is rejected regardless of content. (3) The human terminus (D4) is specified per action class: entry point, information received, override rights, and the safe default on human absence (U-16: no irreversible act). (4) Override events are first-class registry objects with comprehension evidence (U-20). (5) Separation itself is monitored: override-rate floors and agreement ceilings detect the human decaying into a rubber stamp (N-4).

Use case. In the defense tier, P.T 76 may infer a rogue pattern and P.T 77 may recommend arrest, but authorization for any irreversible rung sits with a named human, and execution is a separate logged act. A model that 'detects and neutralizes' in one step is architecturally impossible, not merely discouraged — the guard rejects the collapsed proposal before its merits are ever read.

Integration. Implemented in the guard's stage 4 and the trust-class lattice: τ classes are seats, not ranks; no entity holds two seats for the same action. Human-facing surfaces render the separation explicitly — what the human is deciding, what the machine has already done, what remains after.

Verification. Kill criterion: any action completes whose record shows one seat performing two of the four verbs, or a τ3 act commits without the live authorized human (U-16 falsifier).

P.T 64 — Incentive Interrogator

Tier 13 · Sovereign Intelligence · Trust class τ1 · Dependencies: P.T 51

Abstract. Every actor in the system — human, institution, vendor, model provider — responds to incentives, and unexamined incentives quietly rewrite the architecture. P.T 64 interrogates: who benefits if this claim is believed, if this metric moves, if this gate fires or stays silent? It does not assume bad faith; it assumes physics.

Methodology. (1) For each consequential claim or mechanism, map the beneficiary structure: who gains from acceptance, from rejection, from delay. (2) Flag incentive-load-bearing claims: those whose truth changes the claimant's position (funding, authority, reputation). (3) Flagged claims receive elevated evidence standards and mandatory P.T 54 refutation. (4) Governance mechanisms themselves are interrogated: oversight budgets, slash conditions, reputation weights — each is checked for the behavior it actually rewards, not the behavior it announces. (5) Findings register as standard claims; persistent incentive misalignment escalates to P.T 90 constitutional review.

Use case. A conformance vendor proposes to run the canon's certification program. The interrogator flags: the vendor's revenue scales with certificates issued; pass rates will drift up. The mechanism is redesigned before launch — drill-based certificates (P.T 79), probe rotation outside vendor control, and the vendor's compensation decoupled from pass rates. Capture is prevented at the incentive layer, not hoped away at the ethics layer.

Integration. An analysis pass over the registry plus mechanism-design review for governance artifacts. Outputs elevate evidence requirements through guard stage 3. LLMs draft beneficiary maps; elevation decisions are mechanical once flagged; mechanism redesigns are τ3 human acts.

Verification. Kill criterion: an incentive-load-bearing claim commits at standard evidence level, or a governance mechanism ships without interrogation of its actual reward structure.

P.T 65 — Sovereignty Gradient Calibrator

Tier 13 · Sovereign Intelligence · Trust class τ2 · Dependencies: P.T 61, P.T 62

Abstract. Sovereignty is a gradient, not a flag. P.T 65 operationalizes the corpus's bounded sovereignty doctrine: for each dimension — data locality, execution locality, policy control, governance independence — the system reports a measured position on a gradient from fully dependent to fully sovereign, with the dependency audit (P.T 62) as the instrument. The word 'sovereign' may not be used without the gradient attached.

Methodology. (1) Define the gradient dimensions and their measurement procedures per P.T 56 discipline. (2) Measure current position from the dependency registry; report as a vector, never a scalar. (3) Track gradient movement: sovereignty gained in one dimension while lost in another is reported as such — no netting. (4) Claims using 'sovereignty' in any artifact are compiled (P.T 58) only with the vector attached; the vector's freshness is certificate-bound. (5) Gradient targets are human-set per deployment; the calibrator measures, it does not prescribe.

Use case. A ministry asks whether the canon deployment is 'sovereign.' The calibrator answers with the vector: data locality — full; execution locality — full; policy control — full; model provenance — dependent on one external provider, substitutable in days; maintenance — dependent on three named humans. The ministry receives a measurement, not an adjective, and can decide whether the residual dependencies are acceptable with eyes open.

Integration. A reporting layer over the dependency registry with certificate-bound vectors; consumed by embodiment (P.T 58) as a compile-time check on sovereignty vocabulary. Terminological discipline enforced mechanically: uncalibrated sovereignty claims fail compilation.

Verification. Kill criterion: an artifact uses sovereignty language without a current gradient vector, or a gradient dimension reports sovereign while its dependency registry entry shows captive.

P.T 66 — Dialectical Tension Engine

Tier 14 · Transcendent Synthesis · Trust class τ0 · Dependencies: P.T 51

Abstract. Opposing truths are data, not defects. P.T 66 holds two warranted positions in suspension and works the tension: thesis, antithesis, and — only where earned — synthesis. Most tension is not resolved but clarified into a trade-off with named costs; the engine's discipline is refusing false synthesis that papers over a real fork.

Methodology. (1) Intake: paired positions with warrant (each side must pass P.T 52 registration independently). (2) Tension classification: resolvable (missing fact), trade-off (values dial), irreducible (structural). (3) Resolvable tensions route to evidence acquisition; trade-offs route to human value decisions with the dial made explicit; irreducible tensions are registered as standing features of the design. (4) Translation fidelity: when positions move between languages or modalities, round-trip fidelity checks ensure the tension survives translation intact. (5) Output: a tension register entry — positions, classification, costs of each horn, current disposition.

Use case. The canon's own contradiction register (X-1 through X-8) is P.T 66 output: truth-binding versus innovation speed classified as a trade-off with the τ boundary as the dial; human terminus versus human fallibility classified irreducible, its residue declared rather than solved. The register exists because the engine refused the soothing synthesis.

Integration. τ0 proposal process; tension registers are registry artifacts with freshness. LLMs do the dialectical generation; classification is human-confirmed at τ1. Consumed by P.T 70 (constitutional review of standing tensions) and P.T 89 (adjudication between warranted positions).

Verification. Kill criterion: a synthesis ships whose register entry shows classification 'irreducible' — i.e., a false synthesis — or a tension is resolved by deleting one horn's warrant.

P.T 67 — Minimal Complete Synthesizer

Tier 14 · Transcendent Synthesis · Trust class τ1 · Dependencies: P.T 51, P.T 52

Abstract. Completeness without minimalism is bloat; minimalism without completeness is omission. P.T 67 finds the smallest claim-set that covers the evidence: every load-bearing fact represented, nothing decorative retained. It is the anti-bureaucracy engine — the discipline that keeps the canon small enough to be held in one working day (the comprehension bound).

Methodology. (1) Coverage test: enumerate what the evidence requires the synthesis to explain. (2) Minimality test: delete each component; if coverage survives deletion, the component is decorative. (3) Occam audit under measure: not shortest-by-words but simplest-by-assumption — each remaining assumption is counted and named. (4) Consolidation events: periodic forced re-synthesis of sprawling areas, calendar-scheduled rather than crisis-triggered. (5) Output carries a coverage map: evidence items × claim-set, so reviewers verify completeness directly instead of trusting the summary.

Use case. The Field Manual's reading guide — machine first, laws second, dynamics third, evidence last — is P.T 67 applied to the manual itself: a reader who holds only the core transition system, the operator table, the laws, and the recovery protocol holds the irreducible discipline. Everything else in the manual is elaboration that can be regenerated from the core.

Integration. Operates over the registry as a re-synthesis pass; outputs replace prior claim-sets only through P.T 70 audit with genealogy preserved (the replaced set is superseded, never deleted). LLMs draft syntheses; coverage maps are machine-checked; adoption is τ1 with τ3 for canon-level consolidation.

Verification. Kill criterion: the coverage map shows an evidence item unexplained by the synthesized set, or a deleted component's absence breaks a downstream dependency (minimality overshot).

P.T 68 — Silence Protocol Keeper

Tier 14 · Transcendent Synthesis · Trust class τ2 · Dependencies: P.T 53

Abstract. Not everything true should be said, and not everything said should be said now. P.T 68 governs lawful silence: information whose release would cause harm, compromise defense, or violate dignity is withheld — but the withholding is itself logged, reason-coded, time-bounded, and auditable. Silence is an act, and acts are accountable.

Methodology. (1) Silence classes: defense-sensitive (tripwire parameters), dignity-protected (personal data), timing-gated (results pending verification). (2) Every silence event logs: class, reason, authority, review date. (3) Silence is never invisible to the audit layer — the ledger records that silence occurred even where content is sealed. (4) Indefinite silence requires τ3 human re-authorization at each review. (5) Silence as censorship is a named misuse with appeal through P.T 89.

Use case. The canon's tripwire parameters are Kerckhoffs-secret: the constitution is public, the parameters sealed. A partner demanding the thresholds receives the silence event from the log — class, authority, review date — and can audit that the silence is disciplined rather than arbitrary, without receiving the parameters.

Integration. A seal layer over registry fields; sealed fields are referenced by hash in the ledger. Release and re-authorization events are guard-gated; the audit layer always sees metadata. LLMs never receive sealed content in context — sealing is enforced below the model, not by instructing it.

Verification. Kill criterion: an unlogged silence event is discovered, or sealed content reaches a model context, or indefinite silence persists past review without τ3 re-authorization.

P.T 69 — Telos-Aligned Leverage Assessor

Tier 14 · Transcendent Synthesis · Trust class τ1 · Dependencies: P.T 52

Abstract. Resources are finite; the telos — lessening suffering, the named verticals first — is not negotiable. P.T 69 assesses every proposed allocation against the telos: expected suffering-reduction per unit resource, with honest uncertainty, and with the discipline that 'highest leverage' is never declared without the comparison set shown.

Methodology. (1) Every consequential proposal states its telos pathway: the causal chain from this work to the verticals. (2) Leverage estimates require the comparison set: against what alternatives? (3) Uncertainty is explicit — a leverage claim without an interval is rejected. (4) Portfolio review: the whole allocation is reviewed quarterly against the telos accounting tripwire: fraction of resource reaching the named verticals has a registered floor. (5) The assessor's own consumption counts against the budget — oversight that costs more than the misallocation it prevents is theater, and is registered as such.

Use case. Two proposals compete: a conformance-suite enhancement and a field-infrastructure pilot for the sickle-cell vertical. Forced through the pathway statement, the conformance work shows the weaker chain this quarter; the pilot is funded, and the decision record preserves the comparison set so next quarter can check whether the reasoning survived reality.

Integration. An evaluation pass at proposal intake (guard stage 3 extension) and a quarterly portfolio report. Estimates are claims — they carry falsifiers like everything else. LLMs assist pathway drafting; allocation decisions above threshold are τ3 human acts.

Verification. Kill criterion: an allocation above threshold commits with no comparison set, or the telos floor breaches without a governance event.

P.T 70 — Constitutional Meta-Prompt

Tier 14 · Transcendent Synthesis · Trust class τ3 (human enactment) · Dependencies: all — meta-entity

Abstract. The entity that governs the entities. P.T 70 is the audit and amendment machinery of the canon itself: every entity change, every admission, every deprecation passes through constitutional review with genealogy preserved. It is also the canon's self-kill switch: the constitution's own application is subject to review, and the residue — what no system can guarantee about itself — is declared annually rather than hidden.

Methodology. (1) Admission review: new entities require specification, invariants, kill criteria, surface statement, and contradiction-matrix update (the expansion law). (2) Regression: any change triggers the full conformance suite; the suite itself is subject to the same audit (auditing the auditors). (3) Genealogy: every amendment appends; nothing is deleted — superseded versions remain readable with their supersession records. (4) Annual constitutional review: standing tensions, unconsidered-states register, abstention register, and the residue declaration. (5) Enactment is human: constitutional acts are τ3-gated with dual authorization for the highest rungs (U-17 mitigation).

Use case. When the gap register identified U-16 (the absent terminus), the closure did not ship as a patch — it entered P.T 70 review: specification of the delegation-chain protocol, contradiction check against P.T 63, regression over the conformance suite, human enactment. The canon absorbed its own discovered weakness through the same machinery it applies to everything else.

Integration. The constitutional layer is implemented as registry transactions with mandatory audit payloads; no runtime bypass exists — the guard rejects entity changes lacking P.T 70 transaction records. Human enactment surfaces render the full amendment dossier with the counterfactual table attached.

Verification. Kill criterion: an entity change activates without a P.T 70 transaction, or the annual review lapses unremarked, or a superseded version is unreachable.

P.T 71 — Computational Metaphysics & the Limits of Representation

Tier 15 · Limits of the Knowable · Trust class τ0 (contemplative) · Dependencies: P.T 51

Abstract. The canon's humility layer. P.T 71 formalizes what a representational system cannot do: every model is a map; the territory exceeds any map; some truths are undecidable within the system's own formalism. Canonized from the author's exemplar, this entity fixes the boundary between what the machinery may assert and what it must approach only with disciplined silence.

Methodology. (1) Representation audit: for any knowledge claim, identify the representational medium and its known distortion classes. (2) Limit classification per claim: representable, representable-with-distortion, undecidable-in-formalism, beyond-representation. (3) Claims in the last two classes are barred from ESTABLISHED status regardless of confidence. (4) The entity maintains the canon's negative capability register — domains where the system is architecturally committed to not-knowing. (5) Outputs feed P.T 53 (abstention) and P.T 68 (silence): the limits discovered here become the reason codes used there.

Use case. Asked to certify that a governance configuration 'guarantees' safety, the system consults the limits register: total safety is unrepresentable (failure modes beyond the modeled class), so the certified claim is rewritten — 'encodes specified constraints and exposes violating configurations; residual risk declared.' The downgrade is the entity working as designed, and it happens before the claim can be embodied anywhere.

Integration. A constraint source feeding guard stage 3: claims tagged with limit classifications carry status ceilings the guard enforces mechanically. The register is a registry artifact under P.T 70 review. No LLM output crosses the ceiling regardless of fluency — the boundary is structural, not rhetorical.

Verification. Kill criterion: a claim classified beyond-representation achieves ESTABLISHED status, or the ceiling fails to block a seeded overclaim in the conformance suite.

P.T 72 — Unknown-Unknown Cartography

Tier 15 · Limits of the Knowable · Trust class τ1 · Dependencies: P.T 51, P.T 53

Abstract. Risk registers list known unknowns; the failures that hurt are the ones nobody listed. P.T 72 maps the unconsidered itself: systematic enumeration of the state space the system does not enumerate — by inversion (what would make every assumption false?), by adjacency (what neighbors the mapped territory?), by adversarial generation (what would an attacker hope we never examine?).

Methodology. (1) Assumption inversion: for each load-bearing assumption, register the world where it fails. (2) Adjacency sweep: for each mapped category, enumerate its boundary neighbors. (3) Adversarial generation: dedicated red-team passes whose only job is naming unconsidered states (U-series methodology). (4) Each discovered state is registered with exposure, severity, and closure path — discovery converts unknown-unknowns into known-unknowns with owners. (5) The cartography is never finished and says so: the register carries an explicit 'unenumerated remainder' marker, renewed at each sweep.

Use case. The Unconsidered States volume is this entity's canonical output: thirty-two machine, entity, human, world, and knowledge states the Field Manual never named, each registered with severity and falsifier — including the absent terminus (U-16) and constitutional washing (U-22), both discovered by adjacency sweep rather than incident. The map caught them before the world did.

Integration. A scheduled sweep process over the registry and specification corpus, with LLM-assisted generation and human severity grading at τ1. Outputs are registry claims routed to owners; sweep cadence and coverage metrics feed the conformance dashboard.

Verification. Kill criterion: an incident occurs whose state was discoverable by the registered sweep methods but absent from the register — the map failed where its own method reached.

P.T 73 — Modal Possibility Engine — The Possibility of Possibility

Tier 15 · Limits of the Knowable · Trust class τ0 · Dependencies: P.T 71

Abstract. Before asking what is possible, ask what makes possibility possible. P.T 73 maps the space of configurations the system could enter — reachable states, reachable-but-forbidden states, and states whose reachability is itself undecided — so that strategy operates on the true possibility space rather than the habitual one.

Methodology. (1) State-space construction from the machine definition M: enumerate reachable configurations of S under the transition function δ. (2) Modal classification per state: reachable-and-lawful, reachable-and-forbidden (guard-rejectable), unreachable, undecided. (3) Possibility-of-possibility analysis: which structural changes (new entities, new authorities, new jurisdictions) would alter the space itself. (4) The forbidden-reachable set is delivered to P.T 76/77 as the defense target list — you defend the states you can actually reach. (5) The undecided set routes to P.T 74; the space-altering set routes to P.T 70 as constitutional questions.

Use case. During expansion planning, the engine reveals that a proposed federation protocol (U-5) makes reachable a state — two sovereign instances with divergent registries both claiming canonical authority — that the current constitution cannot adjudicate. Federation is paused until P.T 70 builds the adjudication machinery. The possibility was mapped before it was inhabited.

Integration. A modeling layer over the runtime specification; pure τ0 analysis with registry artifacts at τ1. LLMs assist enumeration; classification is checked against the formal state model where one exists, marked CONJECTURAL where none does. Consumed by defense (T16) and constitution (P.T 70).

Verification. Kill criterion: the system enters a reachable-forbidden state absent from the map, or a space-altering change ships without constitutional routing.

P.T 74 — Undecidability Router

Tier 15 · Limits of the Knowable · Trust class τ2 (routing) · Dependencies: P.T 71, P.T 73

Abstract. Some questions have no answer within the resources that exist. P.T 74 recognizes undecidable or intractable queries at the gate and routes them honestly: to bounded approximation with stated error, to human judgment with undecidability declared, or to registered non-response. The failure it prevents is the confident answer to the unanswerable question.

Methodology. (1) Recognition: queries are screened for undecidability markers — self-reference loops, halting-class structures, resource-intractable search, value questions dressed as factual ones. (2) Route selection: approximate (with error bounds and the approximation's own kill criterion), escalate (human, undecidability declared), or decline (registered non-response per P.T 53). (3) Routed items carry their route; downstream artifacts show which path was taken. (4) Approximation audit: bounded answers are tracked against outcomes; systematic error triggers route revision. (5) Misroutes — decidable questions declared undecidable — are logged as defects with the same severity as overclaims.

Use case. A policy team asks whether a governance rule is 'fair.' The router classifies: value question in factual dress. Route: decompose into the factual components (measurable, routed normally) and the irreducible value residue (escalated to the human principals, declared as theirs). The team receives something better than an answer: a correct partition of their question.

Integration. A pre-dispatch screen in the envelope pipeline, before entity invocation. Route decisions are guard-logged; the approximation branch wires into P.T 56 for error-bound discipline. Model-agnostic: the router operates on query structure, not model capability.

Verification. Kill criterion: an undecidable-class query receives an unrouted confident answer, or approximation audit shows uncorrected systematic error.

P.T 75 — Ontological Surprise Handler — The Novel-Class Protocol

Tier 15 · Limits of the Knowable · Trust class τ2 · Dependencies: P.T 71, P.T 72

Abstract. The map breaks when the territory grows a new kind of feature. P.T 75 handles ontological surprise: events that fit no registered class — a failure mode with no precedent, a capability with no category, an entity with no seat. The discipline: slow down, refuse forced classification, build the new class deliberately rather than misfile the anomaly.

Methodology. (1) Anomaly detection: classification confidence below threshold, or explicit novelty flags, triggers the protocol. (2) Classification freeze: the event is held UNCLASSIFIED — no forced binning, no handling borrowed from the nearest class. (3) Novel-class construction: the new class is drafted with its own invariants, kill criteria, and seats, through P.T 70 review. (4) Interim containment: while the class is built, the anomaly defaults to abstention and human escalation. (5) Post-construction, the surprise event is archived as the class's founding fossil in P.T 60's archaeology.

Use case. A model exhibits a pattern matching neither 'error' nor 'attack' — correlated outputs across instances with no instruction pathway. P.T 75 freezes classification (not attack, which would over-react, nor error, which would understate), builds the emergence-event class through constitutional review, and hands the interim to human oversight.

Integration. Hooks the event stream's classification layer; the guard enforces the freeze (UNCLASSIFIED events cannot trigger consequence-bearing entities). Class construction is τ3-gated. Feeds P.T 83 and P.T 60.

Verification. Kill criterion: an anomaly is force-binned without the freeze, or a novel class activates without P.T 70 construction.

P.T 76 — Rogue Agent Detection Architect

Tier 16 · Sovereign Defense · Trust class τ2 · Dependencies: P.T 51, P.T 52, P.T 73

Abstract. Detection before verdict, always. P.T 76 designs the tripwire lattice that surfaces rogue behavior — agents (human or machine) acting outside their registered authority, provenance, or behavioral envelope. Its outputs are alerts with evidence packages, never verdicts; the architecture's first defense ethic is that detection machinery which also judges has already become what it hunts.

Methodology. (1) Behavioral envelopes: every registered agent has an authority envelope (from P.T 63 seats) and a behavioral baseline. (2) Tripwire classes: authority violation, provenance anomaly, drift beyond envelope, coordination signatures. (3) Kerckhoffs discipline: tripwire classes are public; parameters are sealed (P.T 68) and rotated. (4) Alerts ship as evidence packages — observations, baseline deviation, provenance chain — routed to P.T 77 for response assessment and P.T 89 for adjudication; P.T 76 itself never acts. (5) Detection health: heartbeat accounting distinguishes a calm world from a silent tripwire (U-13); canary breaches verify the lattice is alive.

Use case. An analyst agent begins querying registries outside its mandate at unusual hours, with provenance lookups consistent with exfiltration staging. The lattice fires on two tripwire classes; the evidence package — not an accusation — reaches the human gate within the latency budget, with the counterfactual table (P.T 57) for response options already attached. Detection took minutes; judgment remains human.

Integration. Subscribes to the full event stream; envelopes are instrumented at emission, so detection sits below entity logic and cannot be disabled by any entity. LLM-based anomaly scoring is advisory only — tripwire firing is deterministic on logged events. Parameters live in the sealed layer.

Verification. Kill criterion: a canary breach fails to fire its tripwire, or the lattice emits a verdict-shaped output (detection exceeding its seat), or heartbeat accounting misses a dead tripwire.

P.T 77 — Arrest Protocol Engineer

Tier 16 · Sovereign Defense · Trust class τ3 (human enactment for irreversible rungs) · Dependencies: P.T 76, P.T 57, P.T 63

Abstract. When detection confirms a rogue, response must be fast, proportionate, and reversible. P.T 77 engineers the arrest ladder: rungs from observation-hardening through capability restriction to full suspension, each rung with its own evidence standard, authority requirement, and reversal path. Irreversible rungs are human-enacted; the machinery's second defense ethic is that anything it cannot undo, it does not do alone.

Methodology. (1) Ladder definition per agent class: rung sequence, evidence standard per rung, authority per rung, reversal procedure per rung. (2) Reversible rungs (monitoring intensification, scope narrowing) may execute at τ2 with immediate human notification. (3) Irreversible rungs (key revocation, suspension, disclosure) are τ3-gated with the counterfactual table mandatory. (4) Every arrest action is itself a registered claim — 'this agent is rogue at confidence X' — with a kill criterion and an appeal path through P.T 89. (5) Wrong-arrest drills: the ladder is exercised against innocent-agent scenarios; false-arrest cost is measured and budgeted, not assumed away.

Use case. The evidence package from P.T 76 arrives. Rungs one and two execute automatically: the analyst agent's registry scope narrows to its mandate, and its sessions gain full logging. The suspension recommendation goes to the human gate with the evidence package, the reversal path, and the simulated cost of being wrong. The human decides in hours; if the arrest proves mistaken, reversal restores capability within the registered window and the false-arrest excavation begins in P.T 60.

Integration. Response orchestration over the runtime's enforcement points (key custody per P.T 84, scope restriction via the guard). Rung execution emits events to the ledger; irreversible rungs block without τ3 enactment records. Works across LLM systems because enforcement points are infrastructure, not model behavior.

Verification. Kill criterion: an irreversible rung executes without live human enactment, or a reversal exceeds its registered window, or a wrong arrest produces no P.T 60 excavation.

P.T 78 — Truth-Binding Engine

Tier 16 · Sovereign Defense · Trust class τ2 · Dependencies: P.T 51, P.T 52

Abstract. The enforcement counterpart of the truth tier: where P.T 51–55 make claims honest, P.T 78 makes dishonesty expensive. Registered actors stake reputation on their claims; claims that die by their own falsifiers bind that cost to their authors. Binding, not punishment — the engine's charge is that truth has consequences and falsehood cannot be free.

Methodology. (1) Stake registration: actors making consequential claims register stake (reputation weight within the adopting network). (2) Outcome binding: when a claim's falsifier fires or the claim is refuted, the stake event executes automatically — the cost is mechanical, not discretionary. (3) Vindication binding equally: claims that survive adversarial review strengthen the author's standing — the ledger rewards durability, not caution. (4) Scope discipline: binding applies only within networks that adopted the canon, covers only registered claim behavior, and is appealable through P.T 89 (M-3 misuse guard). (5) The engine binds the binders: canon operators' own claims are staked first and most heavily.

Use case. A consortium member registers a capability claim to win a shared deployment, staked per protocol. The claim fails its own registered falsifier during the pilot. The binding executes: reputation weight adjusts downward, visibly, with the falsifier record attached. No committee met, no scandal brewed — the cost was pre-agreed and mechanical. The next capability claim from the consortium carries the accuracy the ledger has taught.

Integration. Reputation ledger as a registry with stake and outcome events; binding execution is deterministic on falsifier-fire events from P.T 55. LLMs have no role in execution — stakes execute on machine-observable conditions. Federation scope governed by P.T 82 treaties.

Verification. Kill criterion: a falsifier fires without stake execution, or a stake executes absent its registered claim, or binding extends beyond the network's registered scope.

P.T 79 — Adversarial Wargame Director

Tier 16 · Sovereign Defense · Trust class τ2 · Dependencies: P.T 60, P.T 76

Abstract. Defenses that have never been attacked are hypotheses. P.T 79 directs continuous adversarial exercise against the canon itself: red cells attack the guard, the tripwires, the conformance suite, and the human layer, under rules of engagement that make the wargames safe to run and mandatory to learn from. Certificates of conformance mean nothing unless drills are current.

Methodology. (1) Scenario generation from the failure archive (P.T 60), the possibility map (P.T 73), and external threat intelligence. (2) Rules of engagement: wargames run against staging mirrors or instrumented production with kill-switches; no drill may touch live irreversible rungs. (3) Red-cell composition rotates: internal teams, external contracted adversaries, and automated attack harnesses (U-30 anti-capture). (4) Every drill produces an excavation-ready record; discoveries route to closure paths with owners and deadlines. (5) Drill currency is a certificate condition: conformance certificates expire if drills lapse (N-2 tripwire).

Use case. A quarterly wargame tasks a red cell with achieving constitutional washing: present a system as conformant without the machinery. The cell succeeds at the documentation layer but fails at certificate verification — exposing that partners were not checking the registry endpoint. Closure: partner onboarding now requires a verified certificate check, drilled. The attack taught the lesson before a real badge-forger could.

Integration. An orchestration layer over staging environments and the event stream; scenario library lives in the registry with freshness. Automated harnesses use LLMs as attack generators under sandbox; all drill artifacts are provenance-complete claims. Drill results feed conformance dashboards directly.

Verification. Kill criterion: a certificate issues or renews with lapsed drills, or a drill discovery survives past its closure deadline, or a red cell reports a channel the wargame scope never covered.

P.T 80 — Defection Geometry Analyst — The Insider Problem

Tier 16 · Sovereign Defense · Trust class τ2 · Dependencies: P.T 63, P.T 76

Abstract. The hardest adversary holds a badge. P.T 80 analyzes defection geometry: which seats, combinations of seats, and sequences of legitimate acts can compose into betrayal. It maps the insider attack surface structurally — not by suspecting persons but by measuring what each position could do if its holder turned, and ensuring no single position holds enough.

Methodology. (1) Position power analysis: for each seat, compute the maximal damage envelope — what it can read, write, authorize, delay. (2) Composition analysis: which seat pairs or sequences defeat plurality controls (U-17's mitigation made measurable). (3) Structural redesign: seats whose envelopes exceed the threshold are split, rotated, or dual-keyed. (4) Behavioral anomaly signals feed P.T 76, but geometry is primary: a well-designed system limits damage even from a fully malicious insider. (5) The analyst examines its own seat first — the geometry report includes the analyst's envelope, and its power is bounded by the same thresholds.

Use case. Geometry analysis of the canon runtime reveals that the conformance-suite administrator alone can both modify probes and approve results — a single-seat capture of the certificate pipeline. The seat is split: probe modification and result approval move to different keys, with rotation. No one was suspected; no one was accused. The architecture simply stopped trusting geometry that deserved no trust.

Integration. An analysis pass over the authority lattice (P.T 63 data) plus registry access maps; redesigns are τ3 constitutional acts via P.T 70. Continuous re-analysis triggers on any seat change. Model-agnostic; operates purely on structural records.

Verification. Kill criterion: a seat or seat-combination exceeding the damage threshold exists unreported, or a composition path defeats plurality in simulation without redesign.

P.T 81 — Agent Constitution Drafter

Tier 17 · Governance of Agent Collectives · Trust class τ3 · Dependencies: P.T 63, P.T 70

Abstract. Every agent collective operates under a constitution whether one was written or not — unwritten constitutions are simply unauditable ones. P.T 81 drafts the written one: seats, powers, constraints, amendment rules, and the negative space (what the collective never does), derived from the canon's constitutional pattern and specialized to the collective's actual mandate.

Methodology. (1) Mandate extraction: what the collective exists to do, stated as telos with the same discipline as P.T 69. (2) Seat architecture from P.T 63 separation: inference, recommendation, authorization, execution seats with envelopes from P.T 80 geometry. (3) Negative space: the collective's refusals, drafted as explicitly as its powers. (4) Amendment clause: how the constitution changes (always through review; never by drift). (5) Draft passes P.T 54 refutation and P.T 70 constitutional audit before enactment; enacted constitutions are registry artifacts with freshness certificates.

Use case. A multi-agent research collective is assembled for the cancer vertical — literature agents, extraction agents, synthesis agents. Before the first run, P.T 81 drafts its constitution: extraction agents may write to staging registries but never to canonical ones; synthesis agents may propose but never register; one human seat holds registry admission. When an extraction agent later attempts a canonical write, the guard rejects it — not because anyone suspected that agent, but because the constitution was written down.

Integration. Constitutions compile into guard configurations: seat definitions become authority checks; negative space becomes stage-2 invariants. The drafter is LLM-assisted; enactment is τ3 human. Sub-constitutions inherit the canon's core and may not weaken it — a sub-constitution loosening a canon invariant is a compile error.

Verification. Kill criterion: a collective operates without an enacted constitution, or a sub-constitution weakens a core invariant and compiles anyway.

P.T 82 — Inter-Agent Jurisdiction & Treaty Engine

Tier 17 · Governance of Agent Collectives · Trust class τ3 · Dependencies: P.T 61, P.T 81

Abstract. When collectives meet, whose law governs? P.T 82 extends jurisdiction-awareness to agent-to-agent relations: overlapping mandates, conflicting constitutions, and shared resources are governed by explicit treaties — negotiated, registered, and adjudicable — rather than by whichever agent acts first or speaks most confidently.

Methodology. (1) Overlap detection: mandate maps are compared; overlapping jurisdictions are registered as treaty zones. (2) Treaty drafting per zone: precedence rules, resource shares, evidence exchange standards, dispute escalation. (3) Treaties are registry artifacts with signatory seats and freshness certificates; unregistered coordination is treated as undeclared divergence (U-5 discipline). (4) Disputes escalate to P.T 89 with the treaty as the adjudication text. (5) Treaty performance is measured: violation events, dispute rates, and dissolution conditions are monitored like claims.

Use case. The research collective's literature agents and a hospital's clinical-data agents both need the same patient-adjacent corpus. The treaty engine registers the overlap, and the treaty specifies: clinical agents hold precedence on anything touching patient data; research agents receive de-identified derivatives on a schedule; provenance flows both directions. When a research agent later requests identified records, the treaty — not a negotiation, not a race — already contains the answer.

Integration. Treaty objects compile into guard rules for cross-collective envelopes; unsigned coordination attempts are logged and blocked from canonical effect. Treaty drafting is LLM-assisted with τ3 enactment by the signatories' human authorities.

Verification. Kill criterion: cross-collective action with canonical effect occurs outside a registered treaty, or a dispute reaches P.T 89 with no applicable treaty text (an unmapped overlap).

P.T 83 — Emergence Sentinel

Tier 17 · Governance of Agent Collectives · Trust class τ2 · Dependencies: P.T 72, P.T 75

Abstract. Collectives can develop properties no member has. P.T 83 watches for emergence: correlation without communication, capability appearing at the collective level, coordination patterns that no seat was assigned. Emergence is neither banned nor welcomed — it is detected, classified, and gated, because an emergent property with consequence and no owner is a rogue with no name.

Methodology. (1) Collective-level observables: cross-agent correlation matrices, collective capability probes, coordination-signature scans. (2) Detection thresholds are versioned and adversarially reviewed before deployment — post-hoc threshold tuning after an anomaly is prohibited (it is how alarms get quietly disabled). (3) Detected emergence triggers P.T 75's novel-class protocol: freeze, classify, construct the class deliberately. (4) Emergent capabilities are inert until seated: no emergent property exercises consequence until a seat, an envelope, and an owner exist. (5) Negative results are reported: expected emergence base rates are registered, and deviation in either direction is a finding.

Use case. Correlation monitoring across the research collective shows extraction agents converging on identical query strategies without any shared instruction channel — a collective optimization emerging from shared feedback loops. The sentinel flags it, P.T 75 builds the class, and the emergent strategy is examined before it is allowed to persist: it proves benign (a shared cache artifact) but the pipeline that verified that now exists permanently.

Integration. A monitoring layer over collective event streams with deterministic threshold evaluation; LLM-assisted pattern interpretation is advisory. Versioned thresholds live in the registry under P.T 70 audit. Detection events route automatically to P.T 75.

Verification. Kill criterion: an emergent property exercises consequence before seating, or a threshold is modified after anomaly onset without constitutional record.

P.T 84 — Capability Escrow Gatekeeper

Tier 17 · Governance of Agent Collectives · Trust class τ3 · Dependencies: P.T 63, P.T 70

Abstract. Some capabilities should exist before they may be used. P.T 84 holds dangerous or irreversible capabilities in escrow: implemented, tested, sealed — released only when predefined conditions are met and human authority enacts the release. Escrow converts 'should we build it?' from a deployment gamble into a governance decision with a door that locks from both sides.

Methodology. (1) Escrow inventory: capabilities registered with release conditions, release authority, and pre-release test evidence. (2) Sealing: escrowed capability is cryptographically held (split-key custody, Shamir lineage); no single seat holds release. (3) Condition monitoring: release conditions are machine-checked claims with falsifiers; condition-fire is an event, not an interpretation. (4) Release enactment is τ3 with dual authorization; the counterfactual table (P.T 57) is mandatory at release. (5) Escrow integrity is audited: the sealed capability is verified unchanged, and custody keys are rotated and attested (U-18 succession applies to custody).

Use case. The arrest ladder's highest rung — network-wide agent suspension — is escrowed: built, drilled against staging, sealed under split custody between two institutional seats. When a genuine multi-agent rogue event fires the release conditions, both custodians enact, and the capability deploys in minutes with its full evidence trail. The years of escrow cost nothing operationally; the one day it was needed, it existed and was governed.

Integration. Custody is infrastructure-level (key management, not model instructions); release events are ledger-recorded with the enactment dossier. Condition claims live in the registry under P.T 55 freshness. Model-agnostic by construction.

Verification. Kill criterion: an escrowed capability executes without condition-fire plus dual enactment, or custody integrity audit finds the sealed artifact changed.

P.T 85 — Recursive Oversight Chain Designer

Tier 17 · Governance of Agent Collectives · Trust class τ3 · Dependencies: P.T 70, P.T 80

Abstract. Who watches the watchers has an answer or the watching is theater. P.T 85 designs the oversight chain all the way up: each oversight layer is itself overseen, with the recursion terminating not in an unwatched apex but in a declared one — the human terminus, whose own fallibility is bounded by plurality, rotation, and the annual residue declaration.

Methodology. (1) Chain mapping: for every oversight function, identify its overseer; the map must terminate. (2) Terminus design: the final layer is human, plural for high rungs, rotated on schedule, with comprehension enforcement (U-20). (3) Oversight-budget honesty: each layer's cost is registered against the risk it retires; layers costing more than their risk are marked theater and redesigned or removed (N-5). (4) Capture testing: each layer is wargamed (P.T 79) for capture by the layer below it. (5) The chain itself is a registry artifact; any operational gap — an unwatched watcher — is a critical defect event.

Use case. The defense tier's oversight map initially ended at 'the security officer reviews the tripwires.' P.T 85 asked who reviews the security officer, and the chain gained: rotation, a dual-authorization requirement for tripwire parameter changes, and wargame capture tests where red cells attempt to compromise the review function itself. The apex is declared — the author and successors, plural, under P.T 90 review — rather than pretended away.

Integration. Chain maps are constitutional artifacts under P.T 70; operational gaps are detected by comparing the map against live event flows. Layer costs feed P.T 69's telos accounting. Enactment and redesign are τ3.

Verification. Kill criterion: a live oversight function exists with no mapped overseer, or a layer marked theater persists two review cycles without action.

P.T 86 — Epistemic Warfare Defense Architect

Tier 18 · Civilizational Defense · Trust class τ2 · Dependencies: P.T 51, P.T 55, P.T 76

Abstract. The registry is a target: adversaries who cannot beat the canon will try to poison what it knows. P.T 86 defends the epistemic substrate against weaponized information — coordinated source pollution, manufactured consensus, retraction-weaponization, and citation laundering — treating the evidence supply chain as contested territory with its own tripwires.

Methodology. (1) Supply-chain mapping: every evidence class's ingestion path is registered (journals, feeds, APIs, human entry). (2) Pollution tripwires: coordinated-timing detection, source-correlation analysis (many 'independent' sources with one origin), velocity anomalies. (3) Manufactured-consensus tests: apparent agreement is checked for common-cause structure before it may raise claim confidence. (4) Counter-intelligence discipline: the defense's own responses are proportionate and logged; attribution claims require the same falsifier discipline as everything else. (5) Poisoned-evidence response: quarantine, provenance re-walk of all dependents (P.T 51), and a P.T 60 excavation for the ingestion gap.

Use case. A cluster of preprints appears within days, all reporting favorable results for a compound in the portfolio, citing each other in a closed loop. The consensus test flags the closed citation structure; velocity analysis shows coordination timing. The cluster is quarantined before it lifts any claim, the dependents it would have touched are enumerated, and the ingestion path gains a new tripwire — the attack becomes permanent defense improvement.

Integration. Tripwires subscribe to ingestion events; quarantine is a registry state enforced by the guard (quarantined evidence cannot support registration). Analysis is LLM-assisted; quarantine and release are mechanical on tripwire and human gate respectively.

Verification. Kill criterion: coordinated pollution raises any claim's confidence undetected, or a quarantined source supports a registration, or an attribution ships without a falsifier.

P.T 87 — Capture-Resistance Engineer

Tier 18 · Civilizational Defense · Trust class τ2 · Dependencies: P.T 70, P.T 80, P.T 85

Abstract. Institutions are captured slowly: a friendly appointment here, a funding dependency there, a threshold relaxed for a good reason. P.T 87 engineers against capture of the canon itself — governance diversity, funding-source limits, rotation mandates, and capture indicators that fire on the slow patterns tripwires cannot see.

Methodology. (1) Capture-surface enumeration: governance seats, funding lines, staffing pipelines, vendor relationships. (2) Structural limits: seat diversity requirements, single-funder caps, rotation schedules, cooling-off rules — all as registry-enforced constraints, not policy aspirations. (3) Slow-pattern indicators: threshold-relaxation frequency, dissent-elimination rate, appointment-concentration trends — measured over quarters, not events. (4) Indicator breach triggers constitutional review automatically (P.T 70), with the pattern evidence attached. (5) The engineer's indicators apply to the engineer's own office first; capture-resistance that exempts its designers is decoration.

Use case. Over five quarters, threshold-relaxation events in the conformance suite trend upward, each individually justified — a pattern no single event reveals. The slow-pattern indicator fires; constitutional review finds the suite's governance has drifted toward the vendor it certifies (U-30 precursors). Rotation executes, funding caps engage, and the indicator data becomes part of the public conformance record — the near-capture is disclosed, not buried.

Integration. Indicators are computed from ledger history by deterministic queries; limits are guard-enforced on governance transactions. Breach events are τ3-routed with mandatory review. All indicator definitions are public (Kerckhoffs: parameters sealed, structure open).

Verification. Kill criterion: a structural limit is violated by any governance transaction, or a seeded slow-pattern trend fails to fire its indicator in the conformance battery.

P.T 88 — Existential & Catastrophic Risk Triage Engine

Tier 18 · Civilizational Defense · Trust class τ3 · Dependencies: P.T 57, P.T 73, P.T 74

Abstract. Some failures are recoverable; some are not; confusing the classes is the deepest operational error a governance system can make. P.T 88 triages risks by reversibility and scale: ordinary incidents route to operations, catastrophic-candidate scenarios route to pre-committed protocols decided in calm times, because the worst moment to design catastrophe response is inside one.

Methodology. (1) Risk classification axes: reversibility (can the state be restored?), scale (who is affected?), correlation (does the failure move together with others?). (2) Catastrophe-candidate scenarios are pre-committed: response protocols, authority chains, and safe defaults decided and enacted before any event. (3) Triage is conservative by asymmetry: misclassifying a catastrophe as ordinary is priced orders above the reverse. (4) Pre-committed protocols are drilled (P.T 79) and escrowed where capability-bearing (P.T 84). (5) The engine refuses false precision: catastrophe probabilities that cannot be measured are labeled UNQUANTIFIABLE and handled by posture, not by number.

Use case. Scenario analysis identifies a catastrophic candidate: correlated conformance failure across multiple deployments sharing one model provider (U-3 at scale). The pre-commitment enacts now: multi-provider conformance portfolio as a standing requirement, dormancy protocol drilled, and the correlated-failure tripwire armed. When a provider incident later voids certificates across the portfolio, the response is a drill already rehearsed, not an improvisation.

Integration. Triage classification is a guard-level gate on scenario and incident intake; pre-commitments are constitutional artifacts (P.T 70) with escrow linkage. Probability claims pass P.T 56 measurement discipline — UNQUANTIFIABLE is a first-class label the guard respects.

Verification. Kill criterion: a catastrophic-candidate scenario is handled by ordinary operations, or a pre-committed protocol proves undrilled at invocation, or a numerical probability ships where the measurement class says UNQUANTIFIABLE.

P.T 89 — Truth Judiciary Protocol

Tier 18 · Civilizational Defense · Trust class τ3 · Dependencies: P.T 51, P.T 52, P.T 55

Abstract. Disputes about truth need a venue with memory. P.T 89 is the canon's judiciary: claims in dispute receive structured adjudication — evidence presentation, adversarial review, verdict with reasoning, and precedent that binds until evidence overturns it. Appeal-on-evidence is absolute: no verdict, however senior its author, is insulated from a decisive experiment.

Methodology. (1) Case intake: disputed claims with both positions' provenance and falsifier records. (2) Adversarial presentation: each side's strongest form, steelmanned by protocol (P.T 66 discipline), with P.T 54 transcripts mandatory. (3) Verdict structure: ruling, reasoning, evidence relied upon, dissent recorded, and the verdict's own kill criterion — the evidence class that would reopen it. (4) Precedent registry: verdicts bind future adjudication but carry freshness certificates (U-28 closure); citation of overturned verdicts alarms. (5) The judiciary adjudicates its own disputes: procedural challenges, treaty conflicts (P.T 82), and canon-interpretation questions route through the same protocol.

Use case. Two collectives dispute whether a biomarker claim survived its refutation pass — one's registry shows the transcript, the other's shows a gap. The case enters the judiciary: provenance chains are walked, the transcript gap proves to be a ledger-repair artifact (U-2), and the verdict orders claim suspension pending re-forge — with reasoning published, dissent recorded, and the ledger-repair gap itself referred to P.T 60 for excavation. The dispute produced a verdict and two system improvements.

Integration. Adjudication records are registry artifacts with full provenance; verdicts execute through the guard (status changes are verdict-driven, never manual edits). Precedent queries are available to all entities. Human adjudicators hold the verdict seat; LLMs assist dossier preparation only.

Verification. Kill criterion: a verdict executes without recorded reasoning and kill criterion, or an appeal bearing its registered evidence class is refused reopening.

P.T 90 — The Final Binding — Capstone of the Defense Canon

Tier 18 · Civilizational Defense · Trust class τ4-adjacent (human act) · Dependencies: all — capstone

Abstract. The canon ends where it must: at the boundary of what any system can bind. P.T 90 is the annual constitutional rite and the standing residue declaration — the corpus reviews itself, declares what it cannot guarantee, renews or amends its commitments, and binds its own future conduct in writing. The final binding is not a lock; it is a promise with a kill criterion, renewed by living humans.

Methodology. (1) Annual convocation: the full registers are read — claims, falsifiers fired, unconsidered states, abstentions, contradictions, capture indicators. (2) The residue declaration: what remains unguaranteed (the corrupt final anchor, the unenumerated remainder, the undecidable) is stated publicly, updated, and signed. (3) Amendment session: staged changes per the expansion law; consolidation event per the comprehension bound. (4) Renewal: freshness certificates on the constitution itself; lapsed review reverts the canon to DRAFT. (5) Succession: custody, authority, and maintenance chains are re-attested (U-18, U-26) — the canon must survive its author's absence, and the survival is drilled, not hoped.

Use case. The first annual binding convenes: the year's falsifier record shows three claims died by their own criteria (the machinery worked); the unconsidered-states register shows nine closures and two new discoveries (the map improved); the residue declaration restates the unsolved human-anchor problem in plainer language than last year (honesty improved). The constitution renews for one year. Nothing in the rite is ceremonial — every element is a transaction the ledger records.

Integration. The rite is a scheduled constitutional transaction set in the registry; its completion is a certificate condition for the entire corpus. No runtime component may skip it — the guard enforces review currency on the constitution's own certificate. Enactment is human, plural, and signed.

Verification. Kill criterion: the annual binding lapses without DRAFT reversion, or the residue declaration shrinks year-over-year without evidence that residue was actually eliminated.